

ME 333  
MECHANICS OF MATERIALS

Tutorial1b : Concept of Strain

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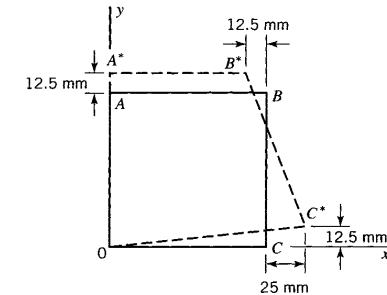
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Tutorial Problems

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[Problem 2.64] A square plate, 1 m long on a side is loaded in a state of plane strain and is deformed as shown.

- 1 Determine the displacements  $(u, v)$ .
- 2 Determine the strains using large strain theory.



Tutorial Problems

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[Problem 2.64] Solution :

Answer : a)  $u = 0.025x - 0.0375xy$ ,  $v = 0.0125(x + y - xy)$ ;  
b)  $\epsilon_{xx} = 0.0254 - 0.0386y + 0.00078y^2$ ,  $\epsilon_{yy} = 0.0126 - 0.0127x + 0.00078x^2$  and  
 $\epsilon_{xy} = 0.0063 + x(-0.0193 + 0.00078y) - 0.0063y$ .

Tutorial Problems

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[Example 1] A two dimensional rectangular bar stretched by uniform end loading results in the following strain field:

$$\epsilon = \begin{bmatrix} A_1 & 0 & 0 \\ 0 & -A_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

where  $A_1$  and  $A_2$  are constants. Assuming the field depends only on  $x$  and  $y$  determine the displacement components and identify rigid rotation and translation terms.

Answer :  $u = A_1x + Cy + D$ ;  $v = -A_2y - Cx + E$ . The constants  $D$  and  $E$  represent rigid translations while the terms going with constant  $C$  are indicative of rigid rotations.

## Tutorial Problems

[Problem 2.67] The strain components in machine parts under small strain theory relative to  $(x, y, z)$  are:

$$\epsilon_{xx} = A(L - x); \quad \epsilon_{yy} = B(L - x); \quad \epsilon_{xy} = 0$$

Determine  $(u, v)$  for the case where the displacement components vanish at  $x = y = 0$  and the slopes  $((\partial u / \partial y), (\partial v / \partial x))$  are equal at  $x = y = 0$ .

Answer :

$$u = A \left( Lx - \frac{x^2}{2} \right) + \frac{1}{2} B y^2; \quad v = B(Ly - xy)$$

## Tutorial Problems

[Example 2] The displacement field for a body is given by:

$$\mathbf{u} = (x^3 + 2y^2) \hat{\mathbf{i}} + (4 + 3z) \hat{\mathbf{j}} + (x + 7y^2) \hat{\mathbf{k}}$$

What is the deformed position of a point that in the undeformed state is at  $(2, -1, 3)$ .

Answer :  $12(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$

## Tutorial Problems

[Example 3] Calculate the distance A'B' given A and B are located at  $(0, 0, 1)$  and  $(2, 0, -1)$ , respectively. Consider the displacement field:

$$\mathbf{u} = (x^3 + 2y^2) \hat{\mathbf{i}} + (4 + 3z) \hat{\mathbf{j}} + (x + 7y^2) \hat{\mathbf{k}}$$

Answer :  $2\sqrt{34}$

## Tutorial Problems

[Example 4] The following state of strain exists at a point A:

$$\boldsymbol{\epsilon} = \begin{bmatrix} 0.02 & -0.06 & 0.03 \\ -0.06 & 0.08 & 0.04 \\ 0.03 & 0.04 & 0 \end{bmatrix}$$

The direction cosines for AB are  $l_1 = 0.8$ ,  $m_1 = 0.6$  and  $n_1 = 0$ . Determine  $\epsilon_{AB}$ . What is the dilatation at point A.

Answer :  $\epsilon_{AB} = -0.016, e = 0.1$

## Tutorial Problems

[Example 5] The following displacement field is imposed on a body:

$$\mathbf{u} = (xy\hat{i} + 3x^2z\hat{j} + 4z\hat{k})10^{-2}$$

Consider a point A and a neighboring point B where AB has the following direction cosines:  $l_1 = 0.2$ ,  $m_1 = 0.8$  and  $n_1 = 0.555$  with A located at  $(2, 1, 3)$ . If AB is  $\Delta s$ , find the components of A'B' post deformation.

Answer :  $(0.218\hat{i} + 0.939\hat{j} + 0.555\hat{k})\Delta s$

## Tutorial Problems

[Example 6] The strain field at a point  $P(x, y, z)$  in an elastic body is given by:

$$\epsilon = \begin{bmatrix} 20 & 3 & 2 \\ 3 & -10 & 5 \\ 2 & 5 & -8 \end{bmatrix} \times 10^{-6}$$

Determine the principal strains and their associated directions. Also calculate the strain invariants.

Answer : Eigen values :  $(20.52, -4.39, -14.13)10^{-6}$ , Corresponding Eigen vectors :  $(-0.9897, -0.1119, -0.089)$ ,  $(0.1394, -0.6161, -0.7753)$ ,  $(-0.0319, 0.7797, -0.6253)$ . The invariants are:  $I_1 = 2 \times 10^{-6}$ ,  $I_2 = 318 \times 10^{-12}$ ,  $I_3 = 1272 \times 10^{-18}$ .

## Reading Assignment

Chapter 2: Section 2.7 - 2.9